AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 4 · STUDENT LESSON

Rules or Learned Patterns?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Choose the rule. Then use it.

Compare two ways to obtain a sorter rule, fit a pebble boundary, practise with visible error counts and try a new length classifier.

Keep one habit: Separate fitting evidence from prediction input and a genuinely new check.

MAIN LESSON · SECTION 1

Who chose the rule, and how?

> Separate writing a rule, fitting a rule and using the finished rule.

Start with the same answer

Two sorters make the same choice. One follows a rule a person wrote; the other uses a rule chosen by comparing labelled examples. The answer alone does not tell you how its rule was obtained.

Retrieve the measurement boundary

A warning uses temperature greater than 28 degrees C. What happens at exactly 28, and what must be checked before comparing a recorded number with 28?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can execute a stated rule including its boundary.
- I can select a rule by comparing errors on **label**led examples.
- I can explain why agreement with **fitting** examples does not establish performance on new cases.

<details>
<summary>Earlier wording, alternative practice and picture transcripts</summary>

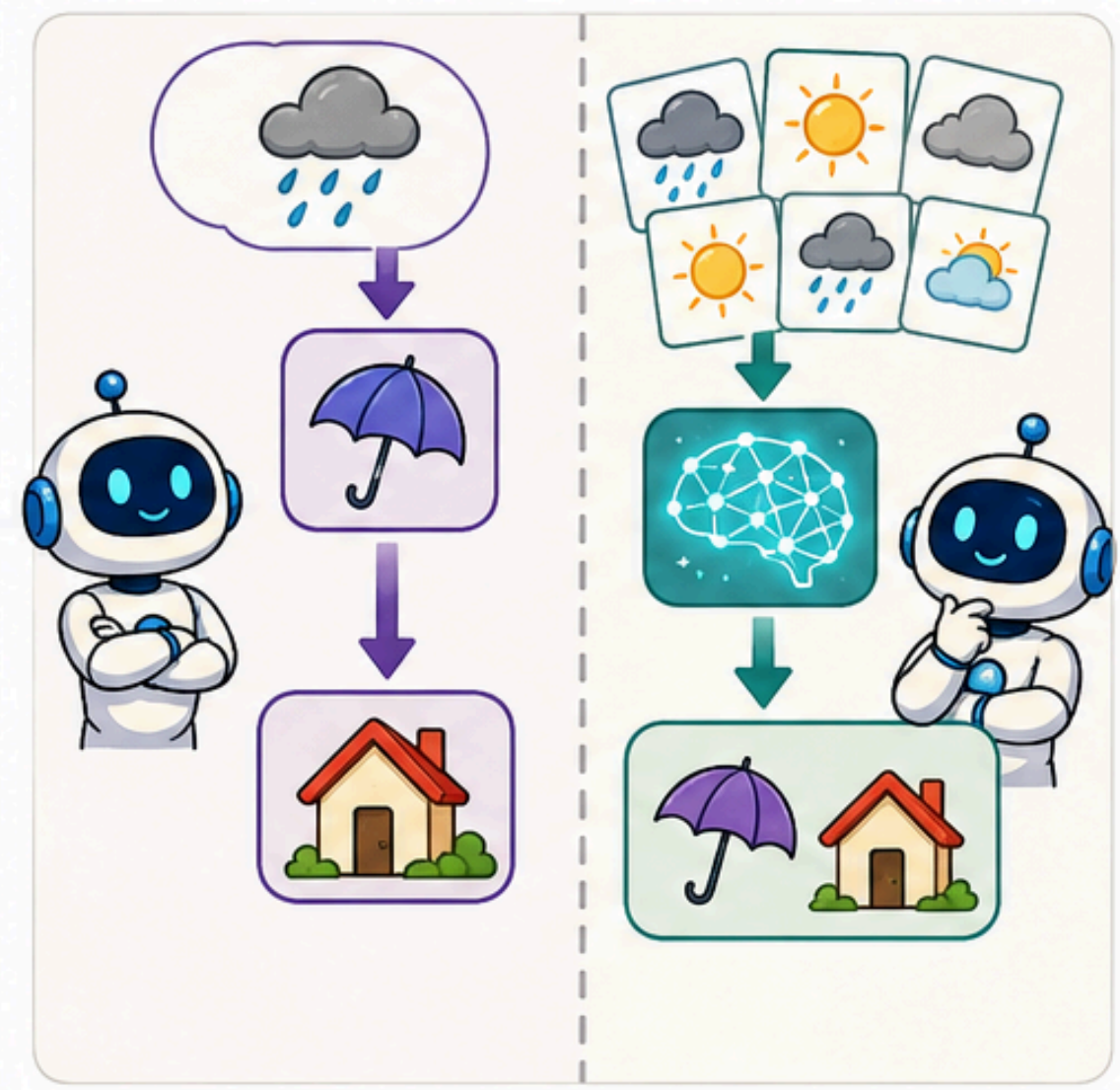
Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

Fixed instructions and learned behavior can look similar but arise differently.
</details>



LEARNING TARGET Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**?

What **evidence** is missing?

Learner name:

Readiness for the connected explanation

This week's end goal

By the end, I can distinguish choosing a rule from applying an already chosen rule.

MAIN LESSON · SECTION 2

Two routes to the same sorting threshold

> Read the task's label convention before comparing rules.

Meet the two paper sorters

A paper sorting task describes imaginary cards. Each card has a width in centimetres and a recorded **label**, narrow or wide. One person writes a rule from a design requirement: **label** a card wide when its width is at least 5 cm.

Another person compares several candidate **threshold**s with **label**led examples and chooses the one with fewest disagreements. Both may end up using the same **threshold**, but they selected it in different ways.

Choosing is different from applying

Executing a rule means applying it to an **input**. **Fitting** means using examples to select or adjust the rule or model. A fitted model can be very simple.

We do not need a brain-like picture or a complicated computer to study the difference between selecting a rule and using it.

Whose labels are these?

In this exercise the **label**s are supplied by the task designer. They are not a universal definition of wide, and a centimetre value alone would not determine every possible meaning of that word.

State what the recorded **label**s mean in the task before counting disagreements.

Describe before you explain

- Name three visible details without explaining them yet.

- Make a first **prediction** and state the clue that supports it.

▼ Earlier wording, alternative practice and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A motion light and an animal recognizer both react to **input**, but one checks a written condition while the other uses patterns learned from examples.

What do you notice?

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Fit first. Predict later. Check afterwards.

> Name which information is available at each stage.

Follow the three stages

How a threshold is selected and then applied

INFORMATION AVAILABLE	OPERATION	RESULT
Widths with supplied narrow or wide labels	Compare candidate rules by error count	Select a threshold
Selected threshold and a new width	Apply the at least comparison	Predicted label
Prediction and a later comparison label	Check for agreement	Evidence about this case

The comparison label is available during fitting and later checking. A new prediction must be made before revealing that case's label.

At least includes the boundary

For **threshold** t, our rule predicts wide when width is at least t and narrow otherwise. At least includes equality: with t=5, width 5 is wide. Changing at least to greater than changes the rule and may change the **error count**.

People choose the fitting setup

A **fitting** procedure also contains choices people made: the **feature**, candidate **threshold**s, **label** definition and way of counting errors. Calling a **threshold** learned does not remove human responsibility for those choices.

Keep the AI distinction precise

Scope note: this course mainly studies learning from examples. AI also includes symbolic and rule-based approaches. Ask how a system reasons or learns; a robot body or a simple automatic switch is insufficient evidence.

Explain an arrow

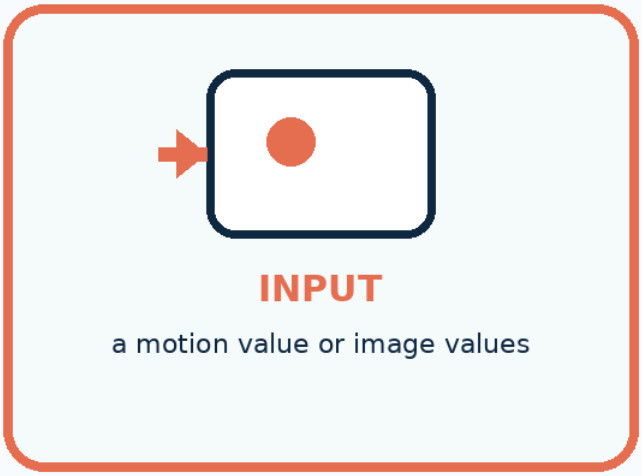
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

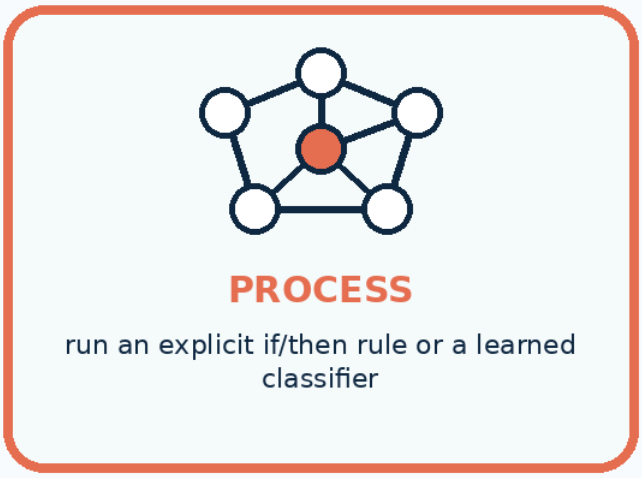
▼ Earlier wording, alternative practice and picture transcripts

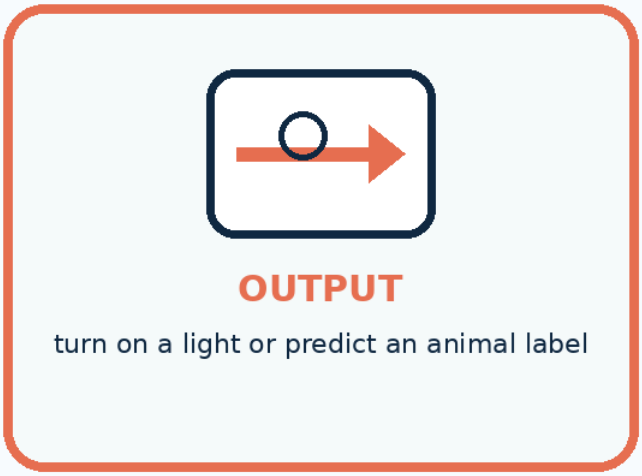
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT a motion value or image values

PROCESS run an explicit if/then rule or a learned classifier

OUTPUT turn on a light or predict an animal **label**

CORE IDEA Fixed instructions and learned behavior can look similar but arise differently.

UNDER THE HOOD A fixed program's decision logic is specified directly by people. A **learned model**'s parameters are fitted from examples.

Both are software, both may be complex, and both can fail; the distinction is how the decision behavior was produced.

Draw a fresh example with all three stages:

A closer explanation

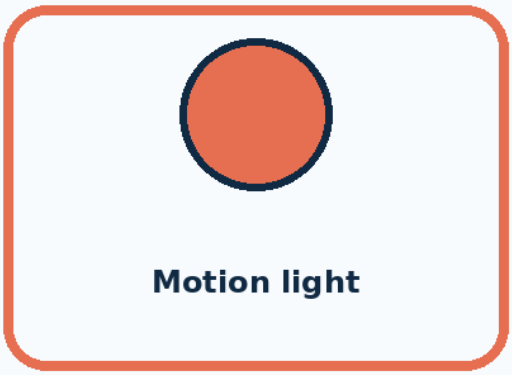
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare evidence about the rule’s origin

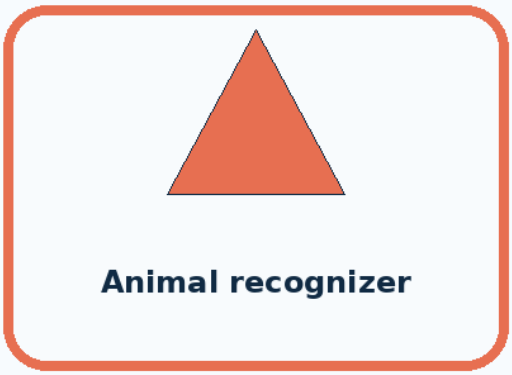
> Simple and complicated do not tell you whether a rule was fitted.

Four cases, paired with their evidence



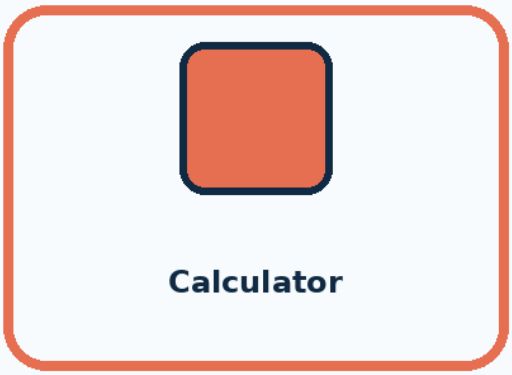
The case to investigate

CASE 1 — Motion light Clue: IF motion, THEN light on. Meaning: **fixed rule**.



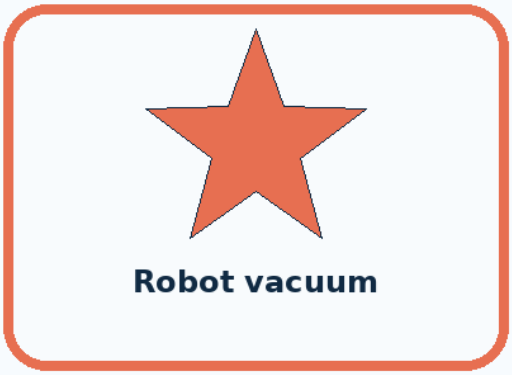
The case to investigate

CASE 2 — Animal recognizer Clue: trained on **label**ed images. Meaning: **learned pattern**.



The case to investigate

CASE 3 — Calculator Clue: follows arithmetic procedures. Meaning: fixed even when complex.



The case to investigate

CASE 4 — Robot vacuum Clue: behavior alone is incomplete **evidence**. Meaning: could be fixed, learned, or mixed.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the distinction clear

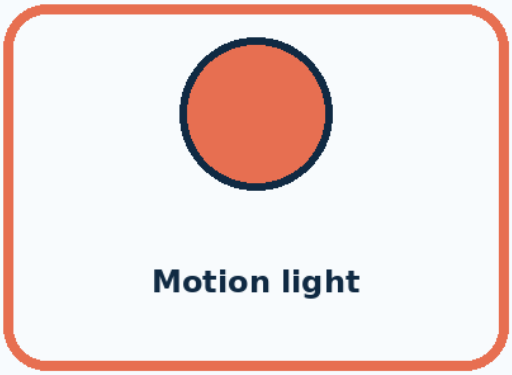
BOUNDARY Do not use complicated versus simple as a substitute for fixed versus learned.

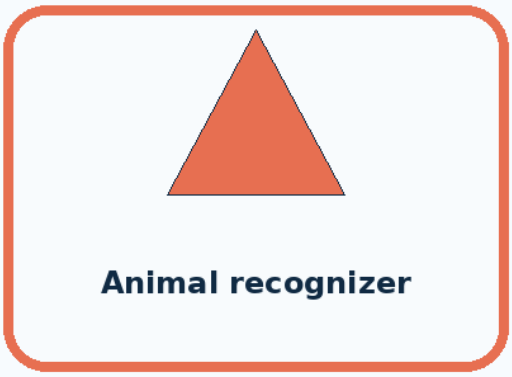
▼ **Earlier wording, alternative practice and picture transcripts**

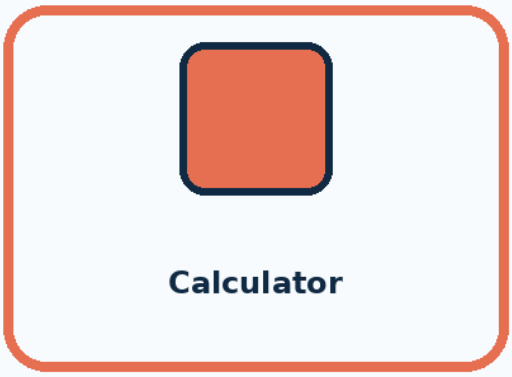
PICTURE GALLERY

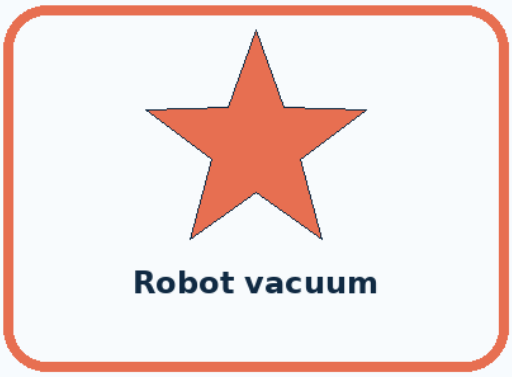
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Motion light Clue: IF motion, THEN light on. Meaning: **fixed rule**.

CASE 2 — Animal recognizer Clue: trained on **label**ed images. Meaning: **learned pattern**.

CASE 3 — Calculator Clue: follows arithmetic procedures. Meaning: fixed even when complex.

CASE 4 — Robot vacuum Clue: behavior alone is incomplete **evidence**. Meaning: could be fixed, learned, or mixed.

Change one thing in the pictured case

Change: Give a new pebble diameter 5 instead of 4.

Prediction: The selected rule predicts large.

Reason: At least includes equality, so 5 meets the boundary.

MAIN LESSON · SECTION 6

Choose a pebble rule using labelled examples

> Compare the candidates on the supplied labels, then use the chosen rule on a new measurement.

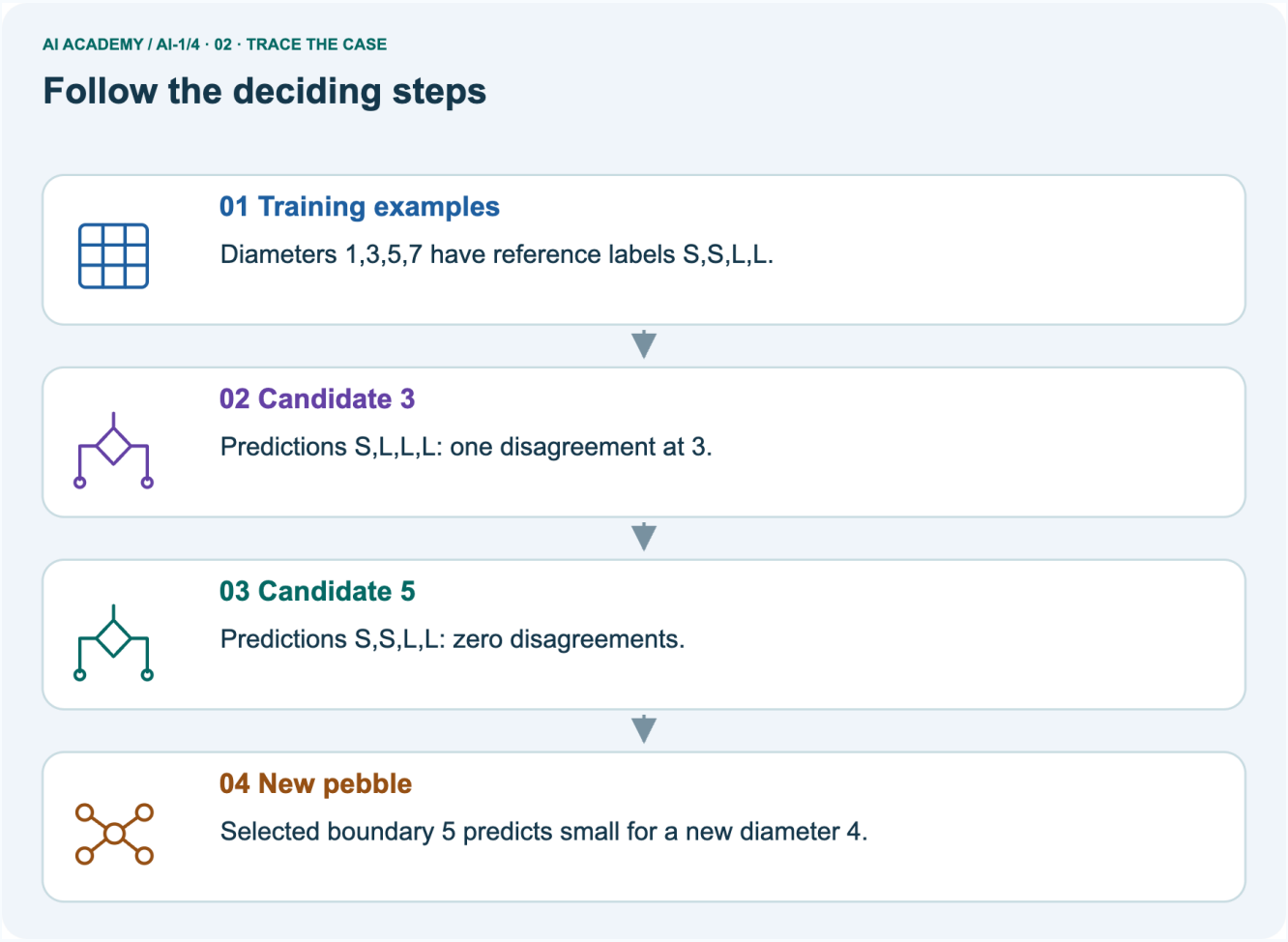
The fictional fitting task

A fictional paper sorter learns a size boundary for toy pebbles.

Training diameters are 1,3,5,7 cm with **label**s small,small,large,large.

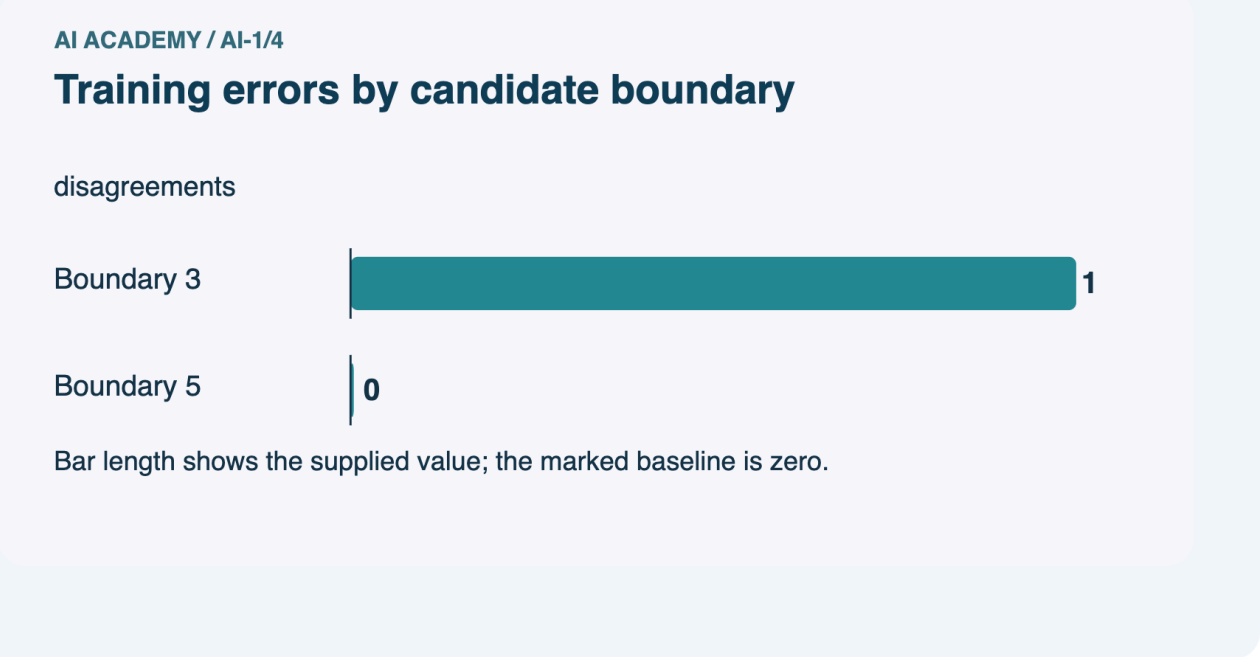
Compare large-at-least-3 with large-at-least-5; choose fewer errors.

Trace selection and the later prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the candidate errors



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why boundary 5 was selected

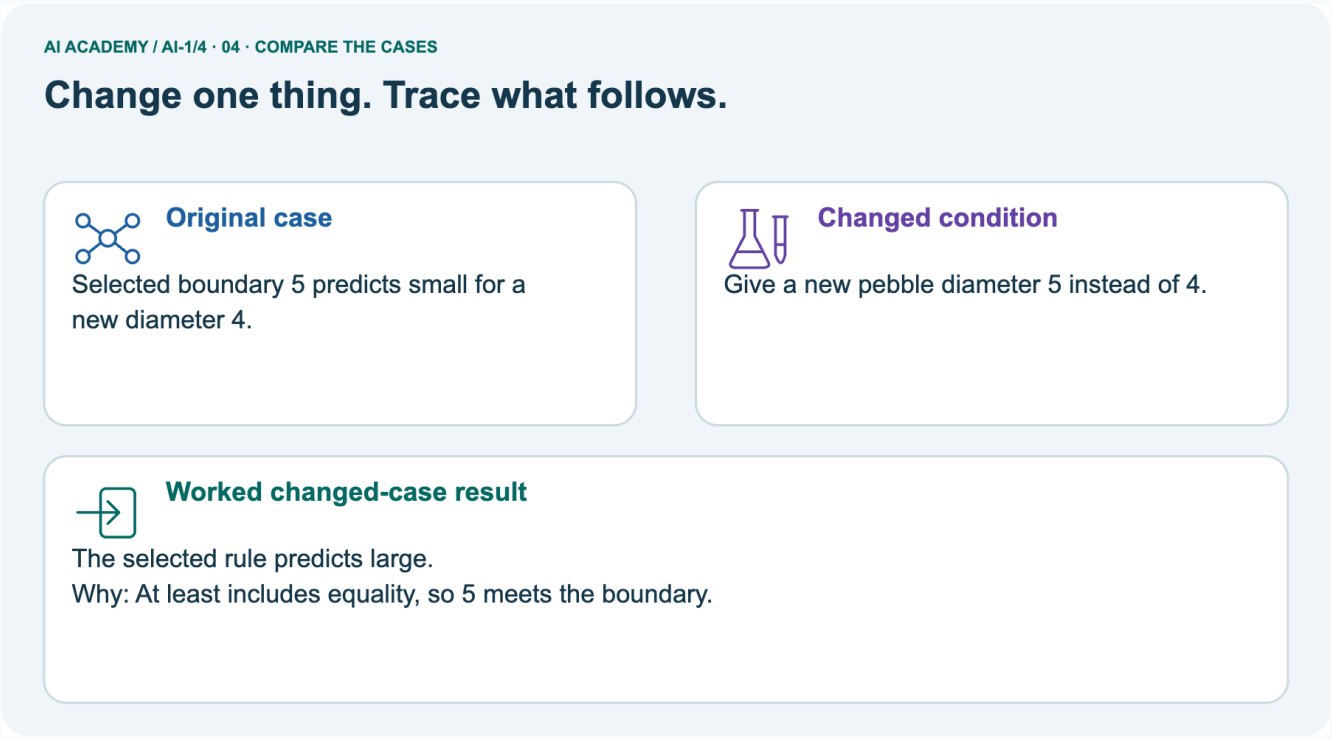
The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

Predict at the boundary

- Keep the selected boundary at 5. Before checking, change only the new pebble's diameter from 4 to 5 cm. Does the prediction change? Did this require fitting the rule again? Explain the equality case.

▼ Check the changed prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

State the limit of zero fitting errors

Zero **training** disagreements do not prove every future pebble receives the correct **label**.

Explain which step used labels

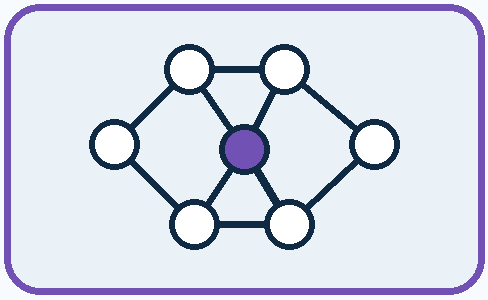
- Read the **label**led picture: Show which stage used **label**s and which stage used only the new measurement.

▼ Earlier wording, alternative practice and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A motion light and an animal recognizer both react to **input**, but one checks a written condition while the other uses patterns learned from examples.

STEP 1 — OBSERVE a motion value or image values

STEP 2 — TRACE run an explicit if/then rule or a learned classifier

STEP 3 — CONCLUDE turn on a light or predict an animal **label**

MODEL REASONING The motion light follows a directly written sensor condition. The animal recogniser uses a model fitted on **label**led examples.

Observing that both react to **input** does not tell us how their decisions were produced.

DO NOT OVERCLAIM Do not use complicated versus simple as a substitute for fixed versus learned.

What new **evidence** could change this answer?

A closer explanation

Width in cm	Supplied label	Threshold 3 prediction	Threshold 5 prediction
2	narrow	narrow	narrow
4	narrow	wide	narrow
6	wide	wide	wide
8	wide	wide	wide

Apply **threshold** 3 to each width before checking the **label**. Only width 4 disagrees, so its **error count** is 1. **Threshold** 5 agrees on all four examples, so its **error count** is 0. Of these two candidates, this procedure selects 5.

Zero disagreements here means it fits these four **label**s; it is not a guarantee about all future cards.

There can be ties. **Threshold** 6 also agrees on these four examples because 6 is included by at least. **Threshold**s 5 and 6 predict differently for width 5, which is absent from the **fitting** set.

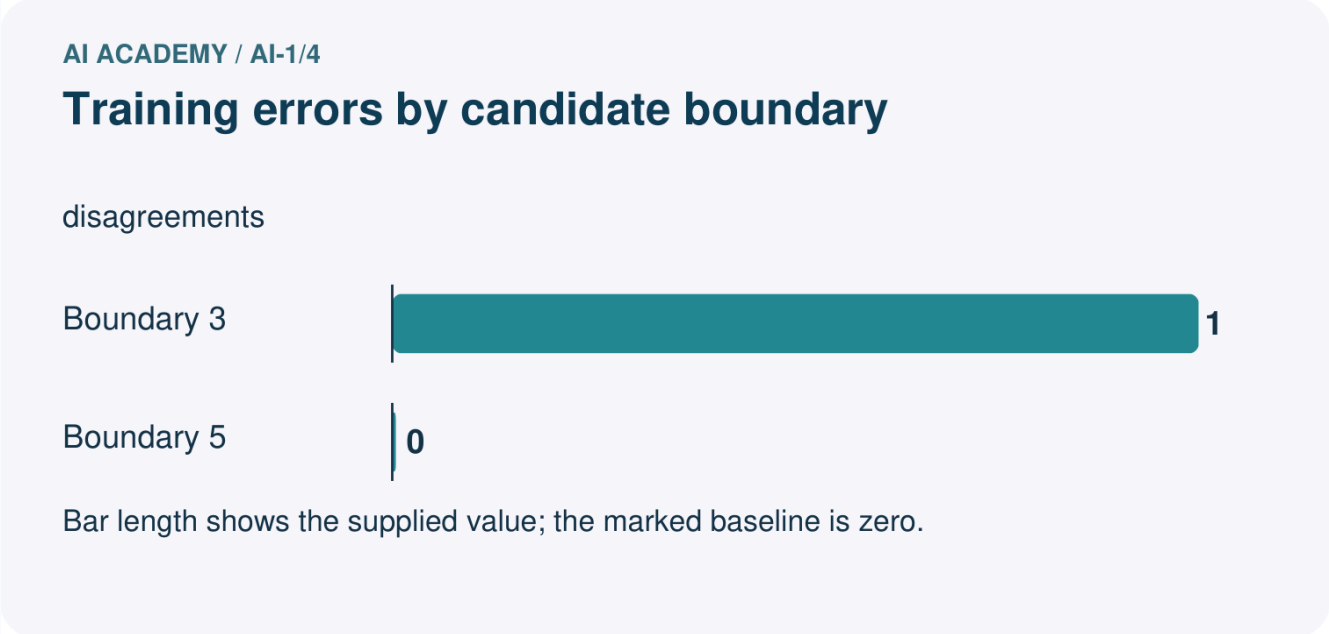
The available examples therefore do not uniquely determine how that width should be **label**led. A procedure needs an explicit tie rule, and we should record the unresolved case.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Rules or Learned Patterns?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/4

Rules or Learned Patterns?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training examples

Diameters 1,3,5,7 have reference labels S,S,L,L.



2. Candidate 3

Predictions S,L,L,L: one disagreement at 3.



3. Candidate 5

Predictions S,S,L,L: zero disagreements.



4. New pebble

Selected boundary 5 predicts small for a new diameter 4.

CHANGE ONE THING

Give a new pebble diameter 5 instead of 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Training examples	Diameters 1,3,5,7 have reference labels S,S,L,L.
2. Candidate 3	Predictions S,L,L,L: one disagreement at 3.
3. Candidate 5	Predictions S,S,L,L: zero disagreements.
4. New pebble	Selected boundary 5 predicts small for a new diameter 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Give a new pebble diameter 5 instead of 4. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Keep a mismatch instead of hiding it

> A fitted rule can disagree with a new supplied label.

Inspect the new-card disagreement

Suppose a new card has width 5 and its supplied task **label** is narrow. The selected **threshold** 5 predicts wide, so it disagrees on this case. Do not erase the example to preserve a perfect score.

Record the error and investigate whether the **label**s follow a different definition, whether measurement matters or whether width alone is insufficient.

Keep a prediction separate from an action

For this week's paper task, a **prediction** is only a proposed category. It does not authorise a physical action.

More consequential applications require additional checks and **human decision**s that a small **fitting** exercise cannot establish.

Repair two tempting claims

- CLAIM A If the **output** changes, the system must be learning.
- CLAIM B A fixed program can never be useful or intelligent-looking.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **Explain the difference between choosing a threshold using examples and applying that threshold to a new card**

▼ **Earlier wording, alternative practice and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Motion light	IF motion, THEN light on	fixed rule
Animal recognizer	trained on labeled images	learned pattern
Calculator	follows arithmetic procedures	fixed even when complex
Robot vacuum	behavior alone is incomplete evidence	could be fixed, learned, or mixed

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: predict every row before counting errors

> Use the W2 comparison as one supported attempt, then explain why an unseen check must stay separate.

The new practice record

For W2, use widths 1, 3, 5 and 7 cm with labels narrow, narrow, wide and wide. For each candidate, predict wide when width is at least the threshold. Write a prediction for every row, then compare it with the supplied label. The size words are the labels of this task, not universal definitions.

Complete and check the comparison

1. Use workbook W2 for the connected example “Rules and **learned pattern**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

> **List the predictions and error count for each candidate. Select the lower-error candidate and state exactly what the result establishes**

Do not turn a final check into fitting data

Keep this comparison separate from a final check on unseen cases. If we repeatedly choose a **threshold** after looking at the check answers, those cases have helped us fit the rule.

They no longer provide an independent check of our finished choice.

Explain and switch roles

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “a learned rule cannot be simple”

- Use the worked case for Rules and **learned pattern**s. Proposed explanation: “A learned rule cannot be written simply”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured label use

Show which stage used **label**s and which stage used only the new measurement.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternative practice and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Fixed instructions and learned behavior can look similar but arise differently.

Use a fresh example to explain **fixed rule**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A fixed program can never be useful or intelligent-looking.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Now use widths 1, 3, 5 and 7 with **label**s narrow, narrow, wide and wide. Compare candidates 3 and 5, using the same at least rule. Write all four **prediction**s for each candidate before you count.

This prevents a plausible-looking total from hiding a boundary mistake.

Practise with less help: Complete the supported comparison

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate: compare the builder’s notes

> Use mechanism evidence to distinguish a written instruction, fitting and an unknown process.

Your supplied case cards

Use these fictional evidence cards. A motion light follows a written instruction: motion=yes → ON, motion=no → OFF, missing reading → request a sensor check. An animal recognizer was fitted using labelled photographs; it displays CAT for one supplied new image. A robot vacuum turns left, but its process notes are missing. These are given facts, not results from a real device test.

Predict, trace and compare

1. Before discussion, predict which cases have evidence of fitting. Trace the motion rule for yes, no and missing. For the recognizer, point to the training fact; CAT alone would not establish how it was obtained or whether it is correct. For the vacuum, write what evidence you would request instead of guessing. Record this as a paper trace and evidence comparison.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep a normal and missing-input record

Case	Prediction	Observed result	What it means
Normal			
Edge			

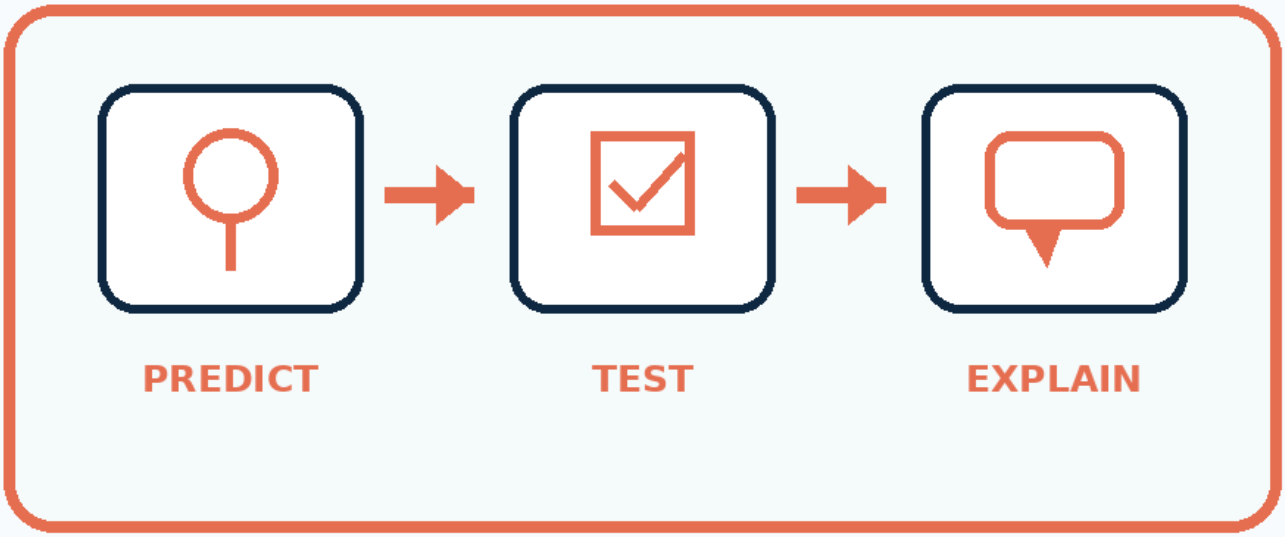
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternative practice and picture transcripts

EXPLORER LAB

Same Behavior, Different Cause

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Compare a motion light with an animal recognizer.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit a different length classifier

> Record both prediction lists and counts, then predict the new length before its label is revealed.

The new case

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

My independent explanation, trace and limits

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Fit a new paper classifier

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

System A uses the **written rule** green means go. System B learns fruit **label**s from example photographs. Who supplies A's rule? What changes when B is trained on different examples?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Record where your sorter’s rule came from

> Keep the written design choice or candidate-error record beside the finished rule.

This week’s project decision

THIS WEEK'S CHECKPOINT Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

Document whether the sorter’s rules were teacher-written or chosen from examples, retaining candidate errors.

Plan evidence and a separate check

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Avoid the complexity shortcut

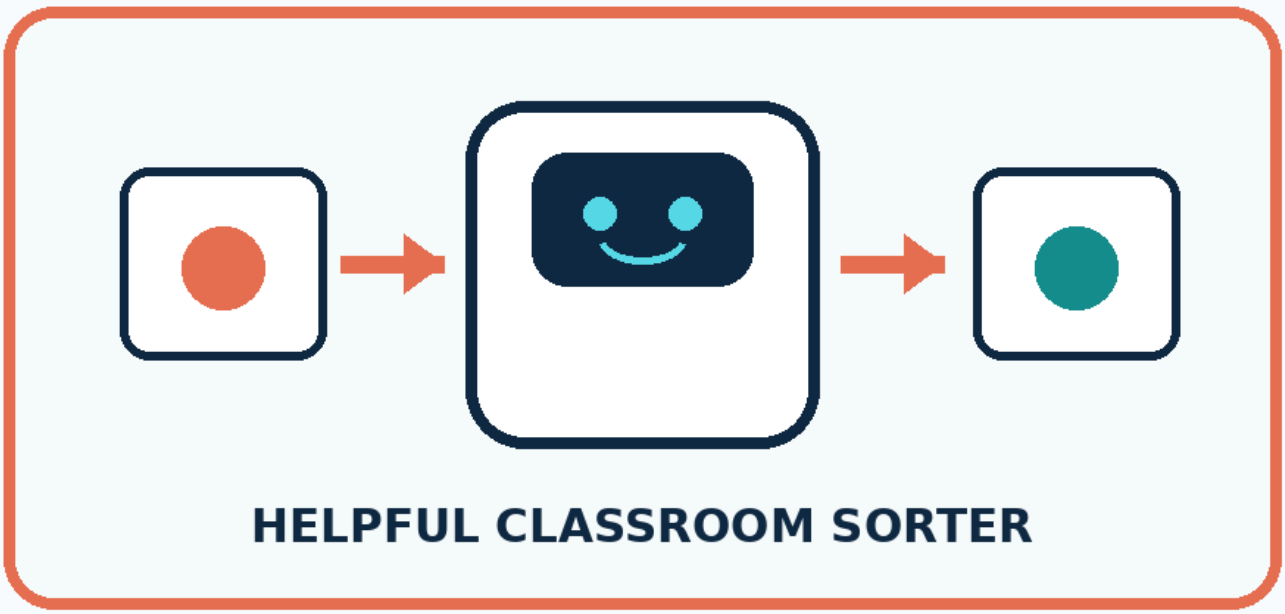
RESPONSIBLE DESIGN Do not use complicated versus simple as a substitute for fixed versus learned.

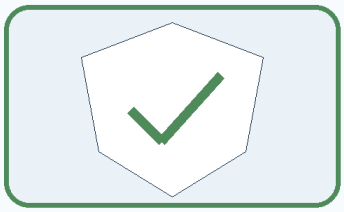
▼ **Earlier wording, alternative practice and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain selection and prediction separately

> Close your notes and say which evidence was used to choose the rule.

Your exit explanation

- Close your notes. Explain this goal in your own words: Execute a stated rule including its boundary

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Choose support and return later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your fitting-and-prediction word guide**

REFERENCE · SECTION 5

Your fitting-and-prediction word guide

> Use the vocabulary to explain a step rather than to label a machine by appearance.

Rule, learning, training and inference

Word	Meaning	My example
fixed rule	instruction written directly in advance	
learned pattern	relationship fitted from examples	
training	process that adjusts a model from examples	
inference	using the finished model on a new input	

Features, labels, thresholds and errors

Word	Meaning
Feature	A recorded property used as an input to a decision or model
Label	A recorded category or target associated with an example
Threshold	A boundary value used in a comparison

Word	Meaning
Fitting	Choosing or adjusting a model using examples
Error count	The number of predictions that disagree with the comparison labels

Repair the changing-output claim

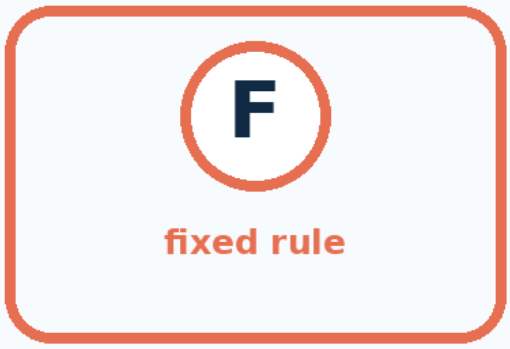
- CLAIM TO REPAIR If the **output** changes, the system must be learning.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternative practice and picture transcripts

VOCABULARY

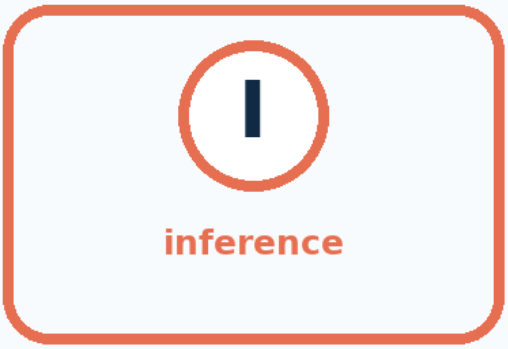
Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger fitting challenge

REFERENCE · SECTION 10

Choose a stronger fitting challenge

> Choose one next step that tests your current explanation.

Pick a useful challenge

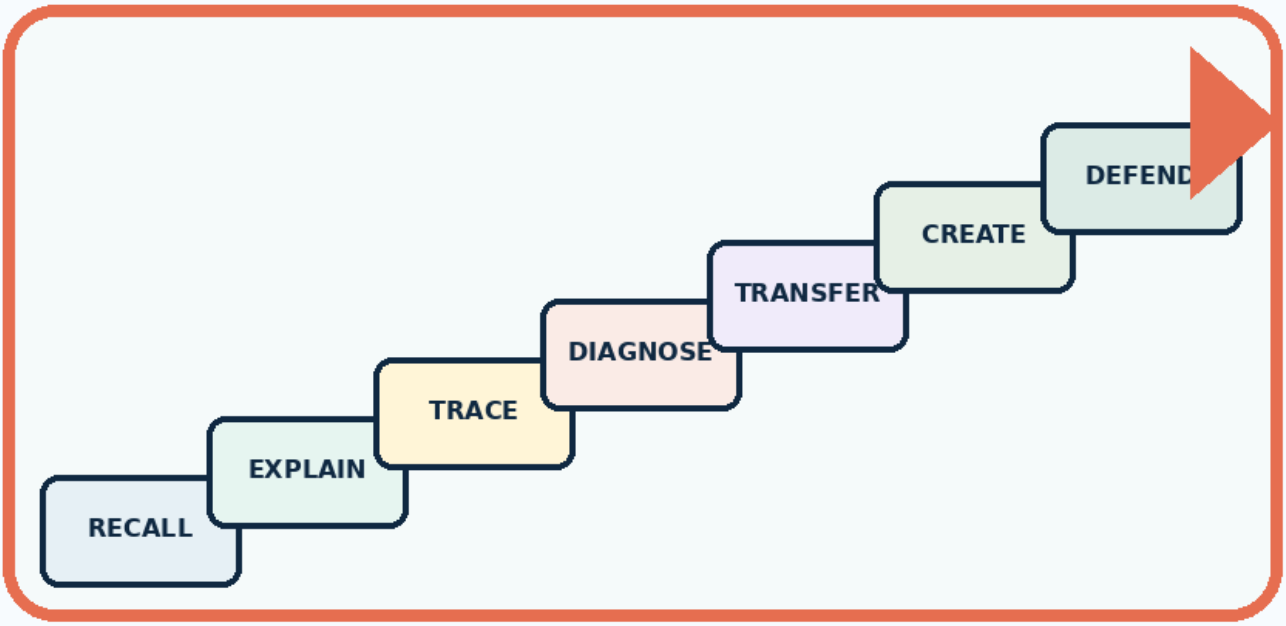
- RECALL Define **fixed rule**.
- EXPLAIN Fixed instructions and learned behavior can look similar but arise differently.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: If the **output** changes, the system must be learning.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternative practice and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check a fitting explanation**

REFERENCE · SECTION 12

Reference: check a fitting explanation

> Use this checklist after practice. The new length-classifier case is the selected independent attempt.

Check the explanation and limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

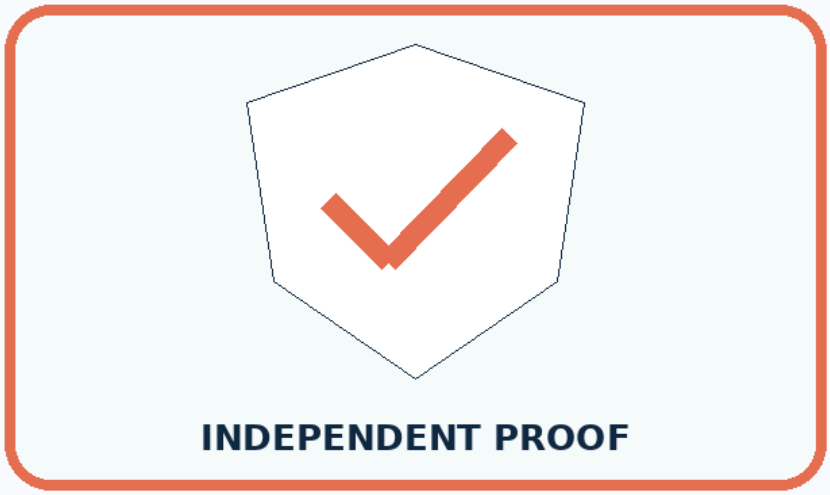
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternative practice and picture transcripts

MASTERY

Checklist for a fitting explanation

Use this checklist after practice. The new length-classifier case is this week's selected independent attempt.



EXPLANATION CHECK Show the candidate predictions, count disagreements with the supplied labels, explain the selected rule and state what remains untested.

Independent **evidence**:

SELF-CHECK The motion light follows a directly written sensor condition. The animal recogniser uses a model fitted on **label**led examples. Observing that both react to **input** does not tell us how their decisions were produced.

NEXT BRIDGE Carry today's **evidence** habit into: Pictures Become Numbers.

▼ **Extra practice: what happens to a blue card?**

REFERENCE · SECTION 13

Extra practice: what happens to a blue card?

> Try the question before opening the worked explanation; do not invent a missing rule.

Read the supplied sorter facts

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

How were their decision **process**es obtained? Where should A send blue cards?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

Evidence or stage	Value or result
Written rule	Red card goes left
Learned pattern	Fit using labelled leaves
Missing rule	Blue-card action unspecified

AI includes more than learning from examples

Scope note: this course mainly studies learning from examples. AI also includes symbolic and rule-based approaches. Ask how a system reasons or learns; a robot body or a simple automatic switch is insufficient **evidence**.

▼ Earlier wording, alternative practice and picture transcripts

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Where did the decision come from?

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

How were their decision **process**es obtained? Where should A send blue cards?

A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist only when it helps inspect your work.

Targets and evidence

By the end, I can	How I will show it
Execute a stated rule including its boundary	Explain the deciding step, show a trace or test, and state one limit.
Select a rule by comparing errors on labelled examples	Explain the deciding step, show a trace or test, and state one limit.
Explain why agreement with fitting examples does not establish performance on new cases	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternative practice and picture transcripts**

Start with this question: A fictional paper sorter learns a size boundary for toy pebbles.

▼ **Extra practice: fitting pictures and feedback**

REFERENCE · SECTION 17

Extra practice: fitting pictures and feedback

> Try before opening each explanation. Keep these supported questions separate from the independent case.

First supported question

- A fictional paper sorter learns a size boundary for toy pebbles.
- Which **label**led step matches this detail? Selected boundary 5 predicts small for a new diameter 4.
- 1. **Training** examples
- 2. Candidate 3
- 3. Candidate 5
- 4. New pebble
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: New pebble Selected boundary 5 predicts small for a new diameter 4. The learning procedure compares candidate rules against supplied examples and selects boundary 5.

Applying that boundary to diameter 4 is a later **prediction** step. An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

Changed-boundary question

- Changed-case practice: Give a new pebble diameter 5 instead of 4. Which conclusion is supported by the supplied facts?
- 1. The selected rule predicts large.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the boundary**

After your attempt, compare with the supported explanation: The selected rule predicts large. At least includes equality, so 5 meets the boundary.

Repair and save

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternative practice and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

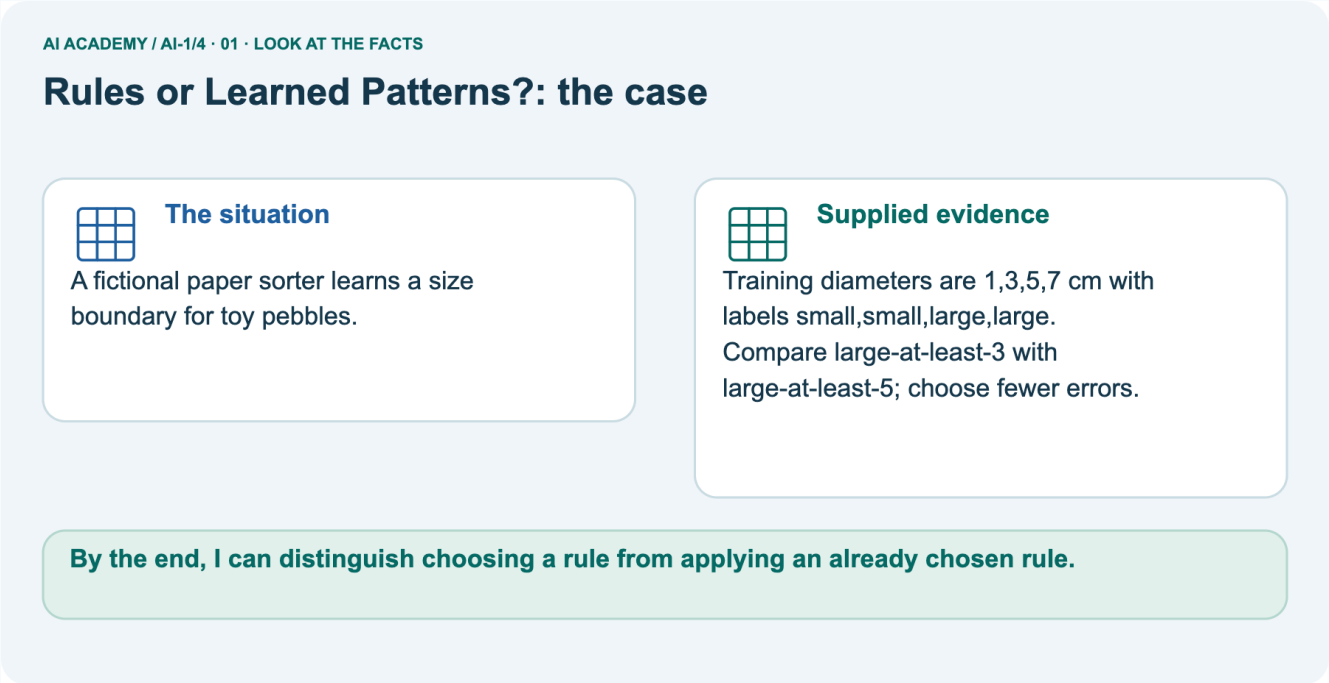
▼ Reference: six-picture fitting journey

REFERENCE · SECTION 18

Reference: six-picture fitting journey

> Use the picture that answers your current question; the main trace is already in the lesson.

01. Rules or Learned Patterns?: the case

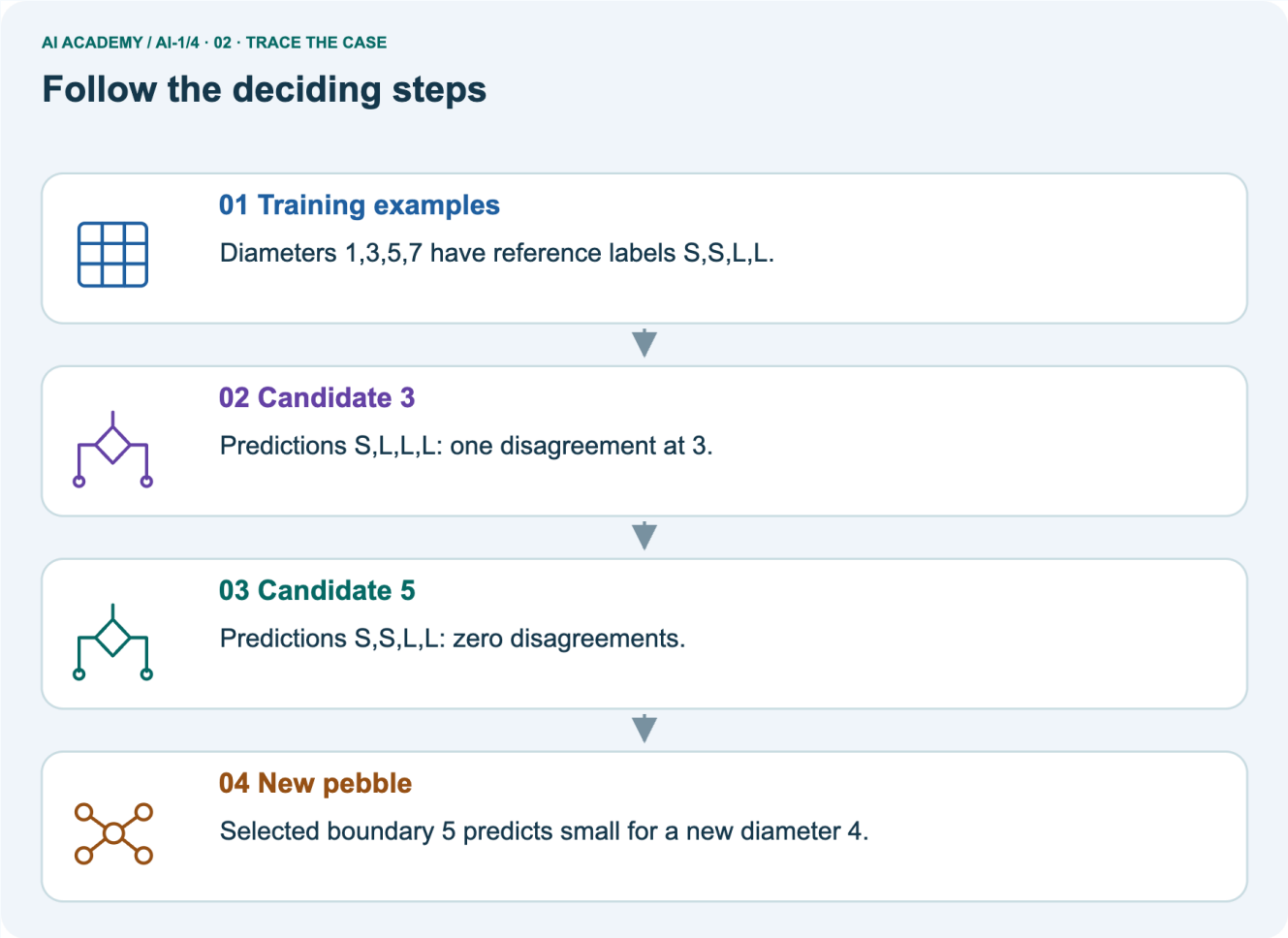


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter learns a size boundary for toy pebbles. Supplied facts: **Training** diameters are 1,3,5,7 cm with **label**s small,small,large,large. Compare large-at-least-3 with large-at-least-5; choose fewer errors.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

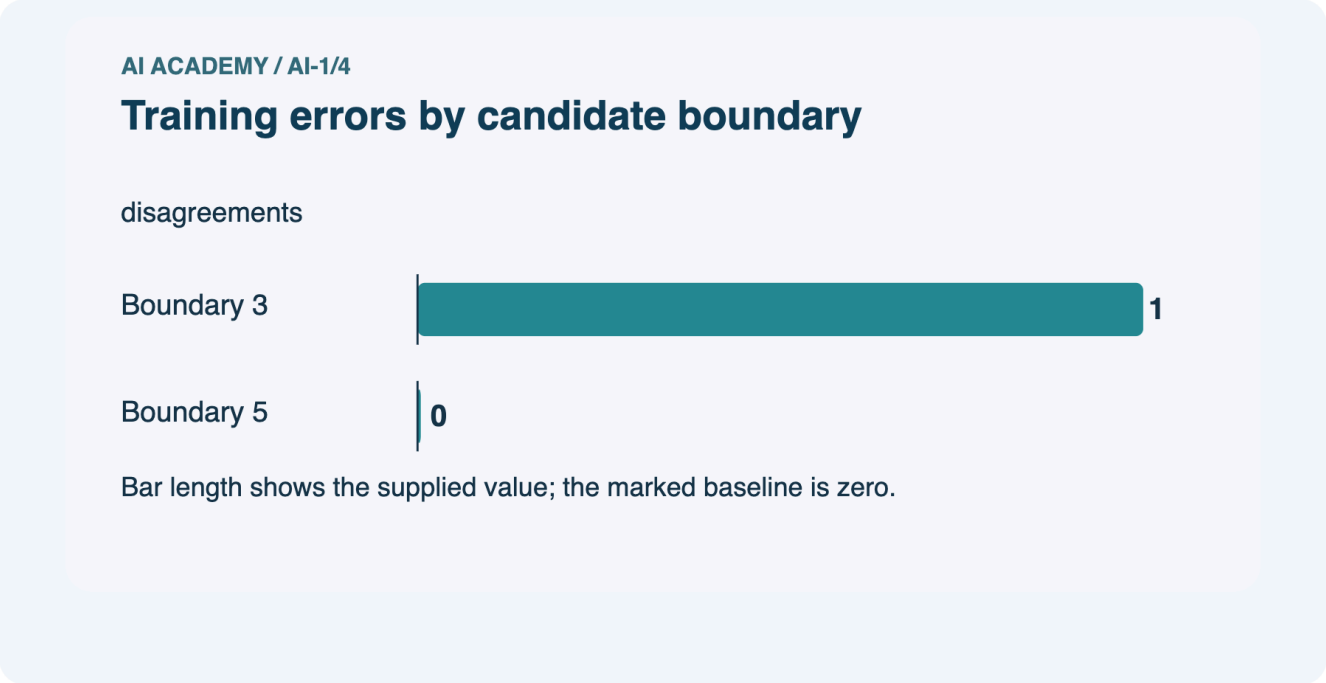


Follow the labelled picture with the matching explanation. Open the vector diagram

Training examples: Diameters 1,3,5,7 have reference **label**s S,S,L,L. → Candidate 3: **Prediction**s S,L,L,L: one disagreement at 3. → Candidate 5: **Prediction**s S,S,L,L: zero disagreements. → New pebble: Selected boundary 5 predicts small for a new diameter 4.

Point to each numbered stage and explain which recorded **input** it uses.

03. Training errors by candidate boundary

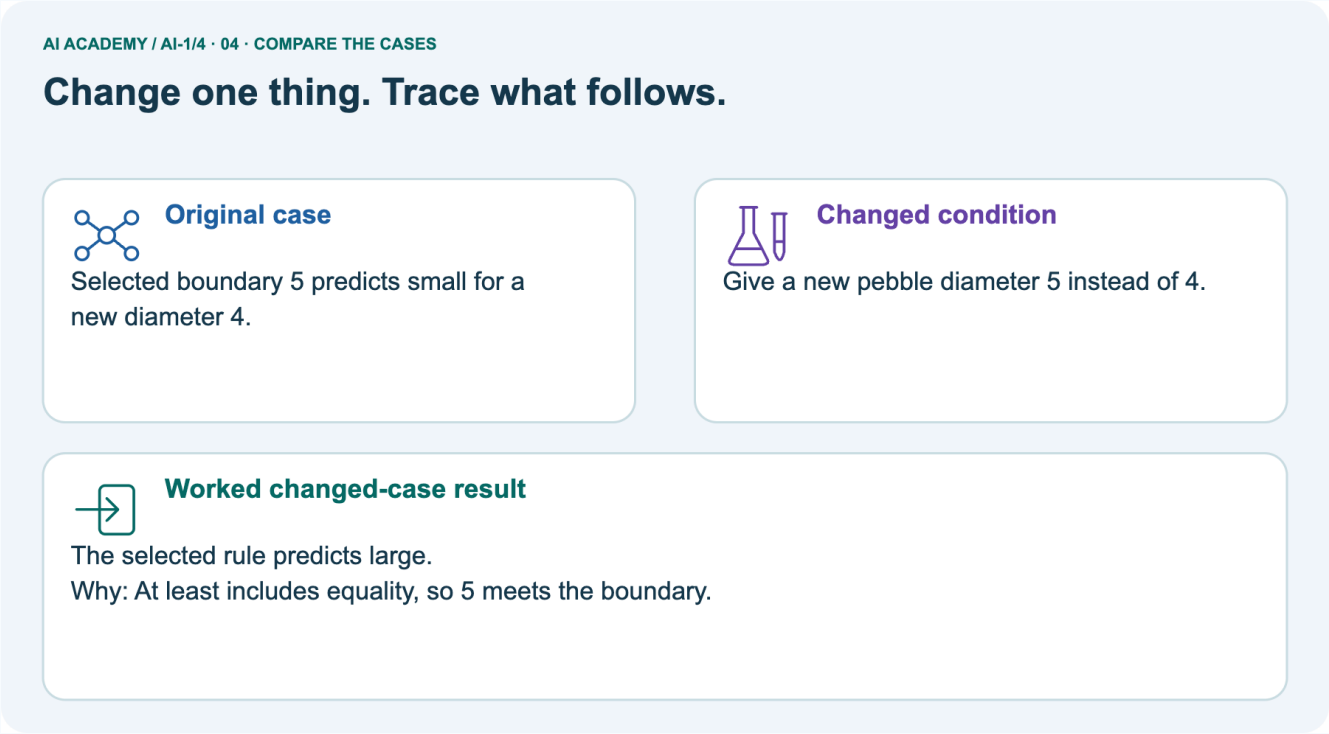


Follow the labelled picture with the matching explanation. Open the vector diagram

Training errors by candidate boundary. Units: disagreements. Boundary 3: 1. Boundary 5: 0.

Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



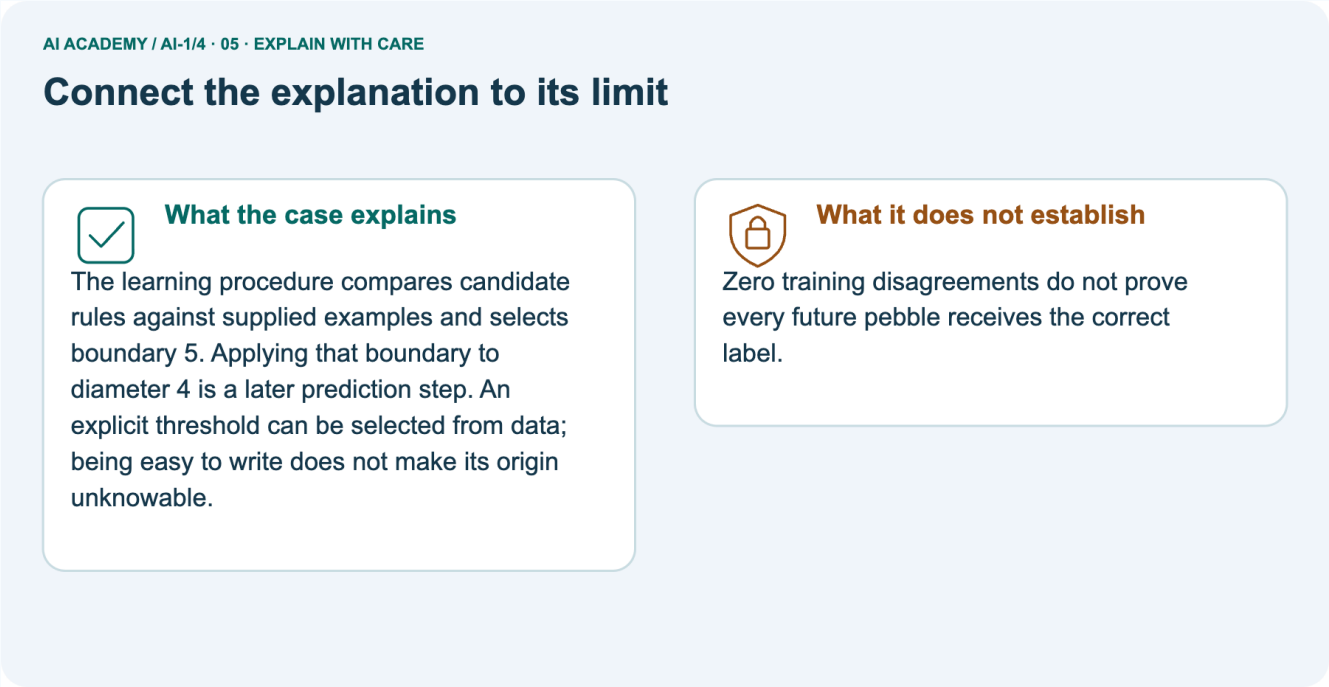
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Selected boundary 5 predicts small for a new diameter 4. Changed condition: Give a new pebble diameter 5 instead of 4. Expected result: The selected rule predicts large.

Reason: At least includes equality, so 5 meets the boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



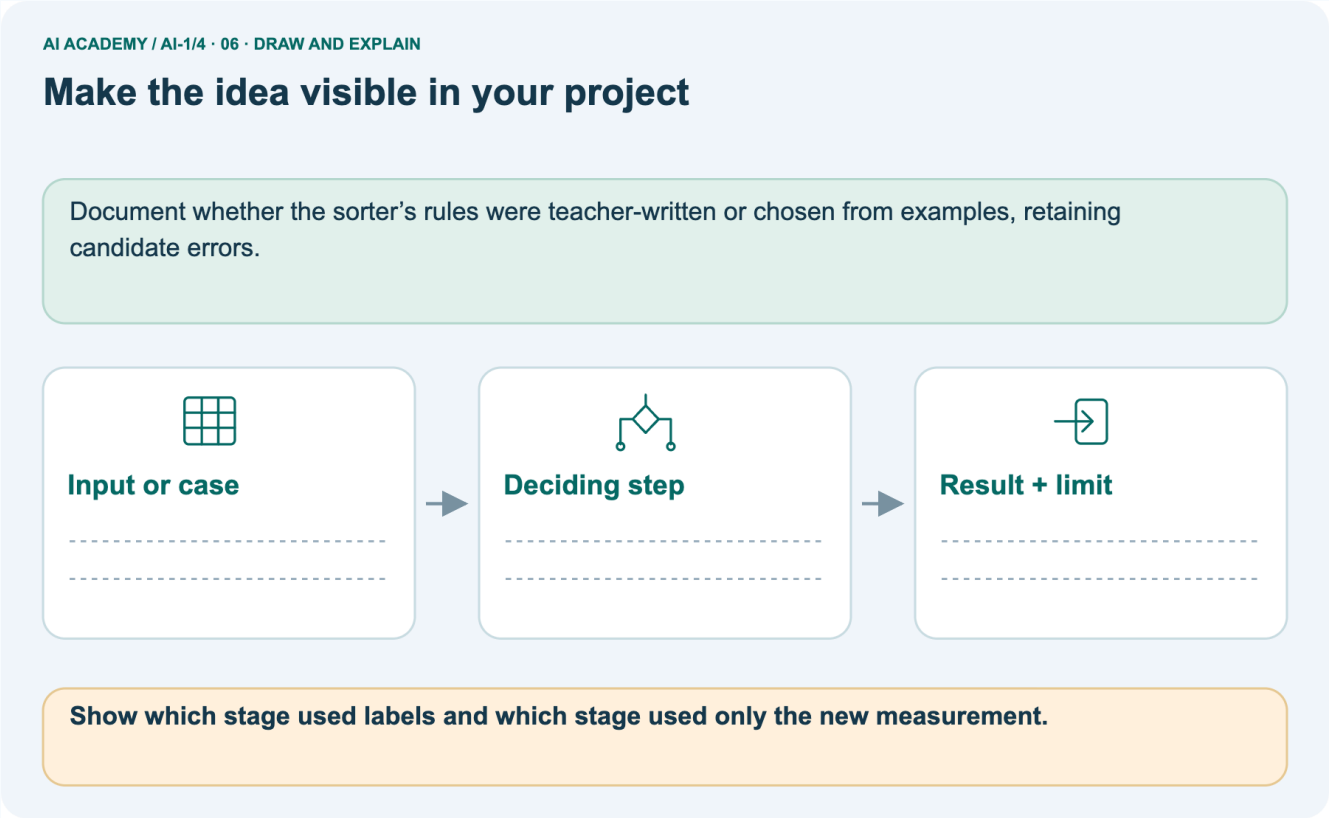
Follow the labelled picture with the matching explanation. Open the vector diagram

The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable. Limit: Zero **training** disagreements do not prove every future pebble receives the correct **label**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors. Draw your **input**, deciding step, result and limit.

Supported question: Show which stage used **label**s and which stage used only the new measurement.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternative practice and picture transcripts

01. Rules or Learned Patterns?: the case

02. Follow the deciding steps

03. Training errors by candidate boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.